

Zhiqian Zhou

Machine Learning · LLM Agents & RAG · Multimodal Models

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Profile

Data Science M.S. student at EPFL working on constrained LLM generation, RAG, multi-agent systems and multimodal models. ICML 2026 workshop co-author; two-time 1st-prize hackathon winner. Native Chinese, bilingual Spanish/Catalan, professional English.

Education

M.S.	EPFL, M.S. in Data Science Focus: Machine Learning and Multimodal Foundation Models	Sep 2026 – Jun 2028 (expected)
B.S.	UPC BarcelonaTech, B.S. in Artificial Intelligence GPA 9/10 (Top 3%) - Distinctions (Top 5%): High Performance Computing, Speech and Dialogue Processing, Knowledge and Automated Reasoning, Systems Modeling and Simulation, Calculus. - Coursework: Algebra, Probability and Statistics, Optimization, Data Management.	Sep 2022 – Jun 2026
Exch.	Beihang University (BUAA), Exchange Student in AI GPA 96/100 - Courses: Foundation Models, Intelligent Security, Medical Image Computing, Image/Video Processing. - HAM10000: DeiT-Small beat ResNet-50 in accuracy, but ResNet-50 led by 0.025 macro-F1 on rare lesions; fetal-ultrasound multimodal fusion; FGSM/PGD robustness tests.	Sep 2025 – Jan 2026

Research Experience

AI Research Assistant Sparsity Technologies & DAMA-UPC	Jul – Sep 2025
- Optimized structured-output generation for LLMs under limited compute. - Built a multimodal knowledge graph of flight and rail schedules to model disruption propagation.	Mar 2026 – May 2026
Bachelor's Thesis, UPC HomeCraft: Text-to-Building Generation in a Voxel World	Feb 2026 – Jun 2026
- Built a multi-agent LLM pipeline that turns one text prompt into a complete voxel building. - Benchmarked open-weight LLMs (CIs, significance tests); fine-tuned a small distilled model.	Advisor: Prof. Ramon Sangüesa

Publication

WinDOM: Self-Family Distillation for Small-Model GUI Grounding. C. Li Chen, Z. Zhou, H. Chen, N. Chauvin. ICML 2026 Second Workshop on Agents in the Wild: Safety, Security, and Beyond. [OpenReview](#)

Key Projects

AlmSafe: Worker Safety Monitoring	2026 1st Prize, MERIThON UPC Repo
Engineered a real-time, privacy-preserving vision system tracking up to 5 workers (fatigue via PERCLOS/gaze, lifting posture, collision risk); fused signals into a REBA safety score with LLM coaching and MQTT equipment control.	Python
OpenCV MediaPipe LLM agent MQTT	
Alvidence: Multi-Agent Misinformation Hunter	2025 1st Prize, MERIThON UPC Repo
Designed a 4-agent LLM and RAG pipeline for fact-checking and credibility scoring, with transparent reasoning logs.	LLM
RAG LangChain Python	
ADDAD: Smadex Creative Intelligence	2026 HackUPC Repo
Architected a low-latency mobile-ad scoring system: soft-voting tabular ensemble (XGBoost, CatBoost, LightGBM), a fine-tuned SmolVLM for structured JSON critiques and a Flux LoRA/DPO pipeline for visual rebuilds, all from a single screenshot.	XGBoost CatBoost LightGBM SmolVLM Diffusion
NextBuy: Smart Demand Signals for Inibsa	2026 3rd Prize, INTERHACK BCN Repo
Built a B2B alert system flagging dental-clinic churn and growth opportunities, ranked by impact with SHAP and Gemini.	
CUPME: Enterprise RAG Chatbot for Siemens Energy	2024 Repo
Engineered a RAG chatbot over Siemens Energy technical docs, using retrieval to minimize hallucinations.	

Honors

Schneider Electric Impact Maker Award 2024 (best mid-degree female AI student) | 5 UPC distinctions (Top 5%) | High School Matrícula de Honor 2021 (GPA 9.8/10)

Skills & Activities

Programming: Python (Advanced), C++, SQL, R, MATLAB | **ML & Data:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas
GenAI & Agents: LangChain, RAG, ReAct, MCP, Tool Use | **Vision & NLP:** CNNs, ViT, OpenCV, MediaPipe
MLOps & Tools: Git, Docker, FastAPI, Weights & Biases, Jira | **Languages:** Chinese (Native), Spanish and Catalan (Bilingual), English (Professional)
Leadership: AECH UPC events committee (2023–2026): 100+ attendee networking events; hosted a Tsinghua delegation
Interests: Boxing, dancing, travel, gastronomy